

Statistics – The **Backbone** of **AI/ML**

Linear Algebra – The Backbone

TOPIC :

Measure of Central Tendency

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CONTENT

Measure of Central Tendency

Mean

Median

Mode

AM

GM

HM

Note:-

Friday I will conduct a

Math Test (Topic - Eigenvalue & vec, C-H Theorem, Vectors)

10X1 = 10 M (on LMS Portal)

① Linear Algebra

② Probability

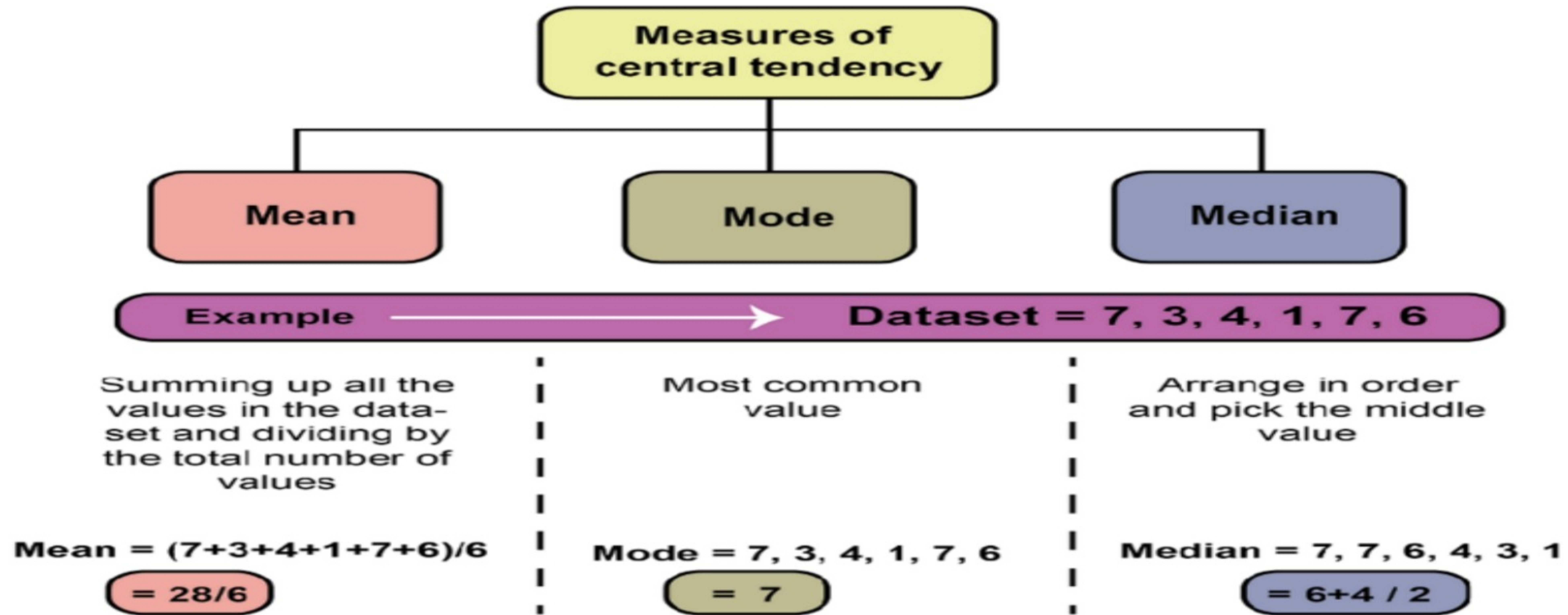
③ Statistics

④ Discrete Mathematics

⑤ Optimization -



The representative value of a data set, generally the central value or the most occurring value that gives a general idea of the whole data set is called Measure of Central Tendency.



Mean

- Arithmetic Mean (AM)
- Geometric Mean (GM)
- Harmonic Mean (HM)

Mean :-
Average

Mean for Ungrouped data

$$\bar{x} = \frac{\sum X_i}{N}$$

OR
Mean = $\frac{\text{Sum of all observation}}{\text{Total no. of observation}}$

Ex:- If there are 5 observations, which are 27, 11, 17, 19 and 21, then the mean ($\bar{x}$) = ?

$$\bar{x} = \frac{95}{5} = 19$$

Mean for grouped data

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} \quad [f_i = \text{frequency}]$$

x_i	4	6	15	10	9
f_i	5	10	8	7	10

Then value of Mean ($\bar{x}$)

$$\bar{x} = \frac{(4 \times 5) + (6 \times 10) + (15 \times 8) + (10 \times 7) + (9 \times 10)}{5 + 10 + 8 + 7 + 10}$$

$$\bar{x} = \frac{360}{40} = 9 \checkmark$$

Arithmetic
Mean

$$\bar{x} = \frac{\sum x_i}{N}$$

where,

$x_1, x_2, x_3, \dots, x_n$ are the observations

N = is the no. of observation

Geometric
Mean

Geometric mean is defined as the n th root of product of " n " number of given dataset.

$$G.M = \sqrt[n]{x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n}$$

Harmonic Mean:

$$HM = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \dots + \frac{1}{x_n}}$$

OR

$$HM = \frac{n}{\sum \left(\frac{1}{x_i} \right)}$$

where, $x_1, x_2, x_3, \dots$ are the observations
and n is the number of observation —

Median:-

Median of Ungrouped data

Case-I when N is odd —

Median = value of observation
at $\left[\frac{(n+1)}{2} \text{th} \right]$ position

Case-II when N is even

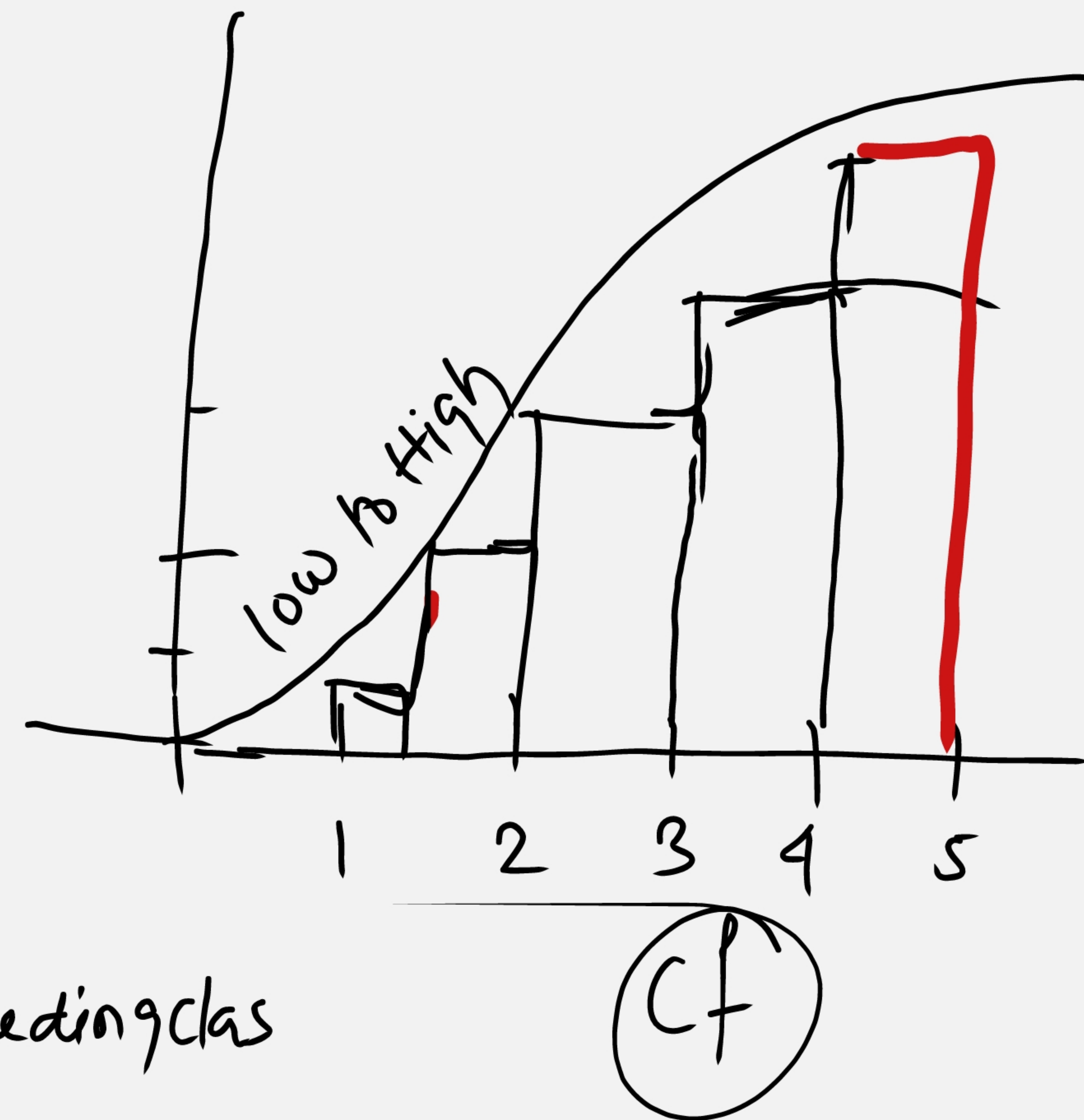
Median = Arithmetic mean of values
of observation at $\left(\frac{n}{2} \right)$ th
and $\left(\frac{n}{2} \right) + 1$ th position —

Median of Grouped data ::

$$\text{Median} = l + \frac{N/2 - C_f}{f} \times h$$

where,

- * l is the lower limit of median class
- * n is the total no. of observations
- * C_f is the Cumulative frequency of the preceding class
- * f is the frequency of each class -
- * h is the class size



Calculate the median^{class} for following data

Class	Frequency	cf
10-20	5	5
20-30	10	15
30-40	12	27
40-50	8	35
50-60	5	40
		40

As, $n = 40$ $n/2 = 20$

30-40 is the median class

$l = 30$, $f = 12$, $cf = 15$, $h = 10$

$$30 + \frac{(20 - 15)}{12} \times 10$$

$$30 + \frac{5}{12} \times 10$$

$$= 30 + \frac{50}{12} = \frac{12 \times 30 + 50}{12}$$

$= 34.16$